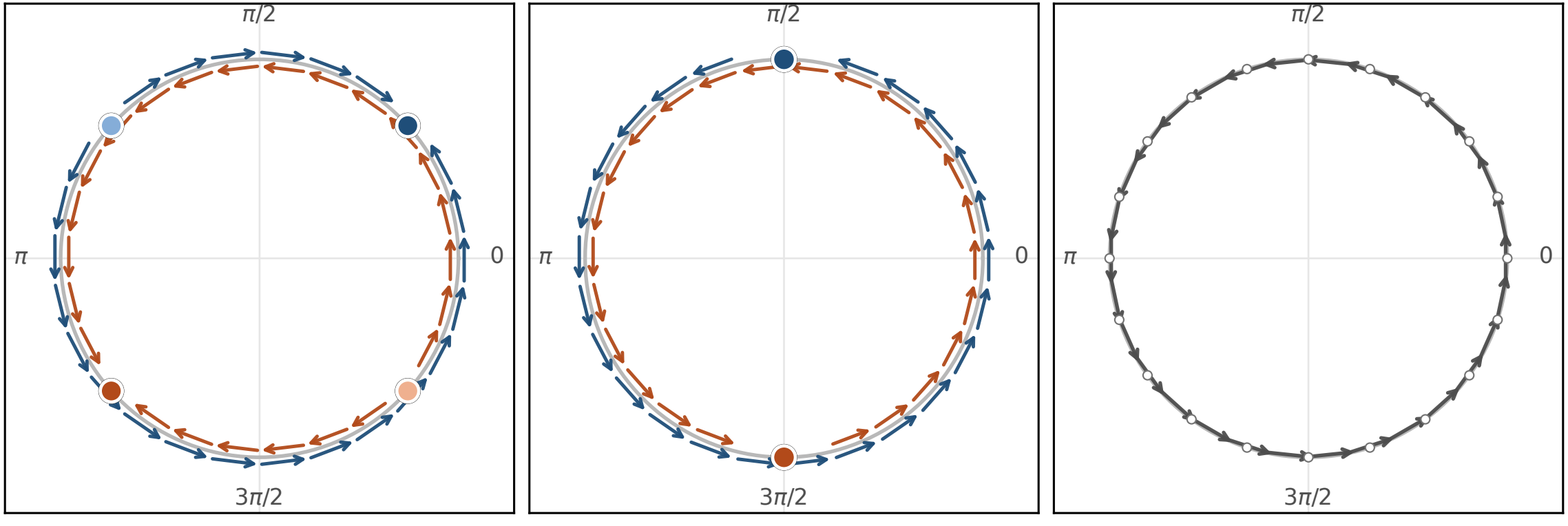
(a) Two fixed points: $|a| > 1$
(b) Critical fixed point: $|a| = 1$
(c) No fixed point: $|a| < 1$
π/2
π
0
3π/2
$a > 1$: Stable FP, Q1
$a > 1$: Unstable FP, Q2
$a < -1$: Stable FP, Q3
$a < -1$: Unstable FP, Q4
$a = 1$: Critical FP at $\pi/2$
$a = -1$: Critical FP at $3\pi/2$